



5. Capstone Project: Full Alert-to-Response Cycle

Objective

The objective of this capstone project was to simulate a real-world cyberattack lifecycle and practice the complete SOC workflow:

1. Attack Simulation
2. Detection
3. Alert Triage
4. Incident Response
5. Containment
6. Reporting and Documentation

Attack Simulation

The attack was performed using the VSFTPD 2.3.4 backdoor exploit available in the Metasploit Framework.

Exploit Used

- Module: exploit/unix/ftp/vsftpd_234_backdoor
- Target Port: 21/TCP
- Vulnerability: VSFTPD Backdoor Command Execution
- MITRE ATT&CK Technique: T1190

Attack Steps:

Step1 — Launch Metasploit

Msfconsole

Step 2 — Select Exploit

use exploit/unix/ftp/vsftpd_234_backdoor

Step 3 — Configure Target

set RHOSTS 10.138.226.251

Step 4 — Execute Exploit

Run

2



Detection and Triage

1	Timestamp	Source IP	Alert Description	MITRE Technique
3	2026-05-14 16:05:12	10.138.226.251	VSFTPD 2.3.4 backdoor exploitation attempt detected	100002
4	2026-05-14 16:06:03	10.138.226.251	Successful exploitation of VSFTPD backdoor	100003
5	2026-05-14 16:06:15	10.138.226.251	Suspicious command execution via FTP	10004

Indicators of Compromise (IOCs)

IOC Type	Value
Attacker IP	10.138.226.251
Target Service	FTP
Target Port	21
Exploit	VSFTPD 2.3.4 Backdoor
Activity	Remote shell access

Incident Analysis

During analysis, it was confirmed that the attacker exploited a publicly accessible FTP service vulnerable to the VSFTPD backdoor vulnerability. The exploit enabled unauthorized shell access to the target machine.

Response:

CrowdSec Mitigation

CrowdSec was configured to block the attacker's IP address.

Command Used

```
sudo cscli decisions add --ip 10.138.226.251 --duration 24h --reason "Attacker blocked"
```



```
root@kali: /home/kali
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# sudo cscli decisions add --ip 10.138.226.251 --duration 24h --reason "Attacker blocked"
INFO Decision successfully added
(root@kali)-[/home/kali]
# cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
2	cscli	Ip:10.138.226.251	Attacker blocked	ban			1	23h59m38s	2

Block IP

```
root@kali: /home/kali
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# ping 10.138.226.251
PING 10.138.226.251 (10.138.226.251) 56(84) bytes of data.
^C
— 10.138.226.251 ping statistics —
63 packets transmitted, 0 received, 100% packet loss, time 64118ms
(root@kali)-[/home/kali]
#
```

Recommendations

- Disable vulnerable and outdated FTP services.
- Apply security patches regularly.
- Restrict public-facing services using firewalls.
- Enable multi-layer monitoring with SIEM solutions.
- Configure automated IPS blocking for repeated attacks.



Incident Response Report

Date: 14 may 2026

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Executive summary:

A simulated cybersecurity incident was conducted in a virtual lab environment using Wazuh, CrowdSec, Kali Linux, and Metasploitable2. The objective was to identify malicious activity, isolate the affected system, and block the attacker IP address. Wazuh generated alerts related to suspicious network activity and possible brute-force attempts. The incident response process was performed successfully.

Timeline

- Attacker machine launched suspicious traffic toward the victim VM.
- Wazuh detected and generated security alerts.
- Logs and alerts were reviewed in the Wazuh dashboard.
- The victim VM network adapter was disabled to isolate the machine.
- CrowdSec was configured to block the attacker IP address.
- Connectivity tests confirmed that communication between attacker and victim was blocked.

Recommendations

- Enable continuous monitoring using Wazuh.
- Regularly update detection rules and signatures.
- Use CrowdSec for automated threat blocking.
- Perform regular vulnerability scans and patch management.
- Maintain incident response documentation and backup procedures.



Stakeholder Briefing:

A cybersecurity incident simulation was completed successfully in the virtual lab environment. Security alerts generated by Wazuh identified suspicious activity originating from the attacker system. Immediate response actions were taken, including isolating the affected virtual machine and blocking the attacker IP address using CrowdSec. Verification tests confirmed that malicious communication was stopped successfully. No permanent damage or data loss occurred during the exercise. The activity demonstrated the effectiveness of monitoring, alerting, and containment procedures. Recommendations include continuous monitoring, regular updates, vulnerability scanning, and improved incident response readiness to strengthen the organization's overall cybersecurity posture.